XUNYANG HONG

R&D Scientist | Scientific Software Developer

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Condensed matter physicist specializing in synchrotron X-ray scattering experiment, combining scientific software development and experimental automation with interdisciplinary problem-solving and visual communication.

Personal webpage: xhong.net | *LinkedIn homepage*

Availability: from Mar. 2027

EDUCATION

PhD in Condensed Matter Physics — Mar. 2023 - Present

Universität Zürich | Group of Prof. Johan Chang

PhD Thesis: Elementary Excitations in Strongly Correlated Superconducting Systems

MSc in Physics — Sept. 2020 - Sept. 2022

ETH Zürich

Master Thesis: Charge and Lattice dynamics in La-based Cuprate Superconductors

BSc in Physics — Sept. 2016 - June 2020

Sun Yat-sen University

Bachelor Thesis: Gravitational Wave in a Cosmological Background

LANGUAGES

English	Proficient - C1
German	Intermediate - B1
Mandarin	Native - C2
Cantonese	Native - C2

CORE STRENGTHS

Experimental Physics:
- 20+ synchrotron X-ray experiments at 4+ international facilities.
- Strong theoretical and mathematical foundation.
- Experienced in physical simulation.

Software Development & Automation:
- Python scientific software and GUI development.
- Website development, maintenance, and workflow automation.
- Solution-oriented: translating practical needs into functional tools.

Adaptability & Emerging Technologies:
- Rapid, independent learning across different disciplines.
- AI integration into research and daily workflows.

Scientific Communication & Visual Design:
- Scientific presentations, visual story telling, and graphic design.
- Skilled in guiding audience attention and simplifying complex concepts

KEY ACHIEVEMENTS

Open-source X-ray software

PyPI-project: developed and published a python application for X-ray calculation and visualization for scattering experiment.

6x faster Laue automation

Developed a Laue diffraction automation combining robot control, reducing procedure time from 30 to 5 minutes, matching manual precision.

Lab digital inventory

Developed a web-base application as a digital lab database keep tracking of items and consumables in the lab.

Streamlined website updates

Developed a web-based reporting system and admin GUI, connecting information collection and website update in one workflow

Four academic papers:

First-author papers on superconducting materials. Combing theoretical models, physics simulations, and experiment analysis

5.9/6.0 evaluation for teaching

7 semesters of teaching experience as teaching assistant. Earning strongly positive student feedback for clear explanations and effective support